

Intelligent Infrastructure Architecture Towards an AI-Driven Automation and Security Framework for the CaTRam Data Center

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Abstract—The modern data center environments increasingly demand intelligent, self-governing operations that go beyond human-centric management at scale. The proposal is based on the live production infrastructure of a Canadian-based transit technology company that will be called 'CaTRam' to preserve anonymity, in accordance with institutional ethics policies.

The underlying infrastructure for CaTRam is a heterogeneous, multi-tier deployment of dual HAProxy load balancers, a three-node Percona XtraDB Galera cluster, a NATS JetStream messaging fabric, KeyDB caching tiers, and five control servers, all of which are CentOS 7 and modern Ubuntu 22.04 hosts. This complexity means that reactive, dashboard-driven operations are not enough. In particular, three control servers (s1, s2, s3) are running CentOS Linux 7 (end-of-life June 2024), and the web tier is still running PHP 7.4-FPM (end-of-life November 2022). The monitoring landscape today consists of tools like PMM, VictoriaMetrics, InfluxDB, Grafana, and Prometheus. This results in a fragmented monitoring landscape with no common intelligence layer. The operators need to correlate signals manually across seventeen nodes.

This proposal outlines a master's thesis project on the design, implementation, and evaluation of CaTRam, an AI-driven automation and security configuration framework. The framework includes anomaly detection based on multi-source telemetry utilising an LSTM autoencoder and provides operational insights via Slack Bolt webhooks and Google Workspace Sheets for audit logging. Remediation actions will be orchestrated via Python FastAPI through Ansible-driven playbooks.

We will experimentally evaluate more than 30 synthetic fault injection scenarios targeting at least a 60% reduction in mean time to recovery (MTTR) and a 75% reduction in false positive rate compared to the manual intervention-based baseline operations.

Keywords — AI-driven automation; anomaly detection; data centre security; Galera cluster; NATS JetStream; LSTM autoencoder; Ansible; ChatOps; CrowdSec; Slack integration; Google Workspace

I. INTRODUCTION

Infrastructure complexity surpasses human cognitive bandwidth. This research takes place within the live production environment of a Canadian-based transit technology company (anonymised in accordance with institutional ethics policy), whose infrastructure is referred to throughout as CaTRam. The CaTRam platform hosts the DaVinci CP application based on the Symfony PHP Framework on 17 nodes on the internal subnet 172.22.64.0/24, on management networks (172.18.x.x and 172.19.x.x), and on several public IP ranges. Dual HAProxy load balancers with Keepalived

VRRP failover (VIP: 172.22.64.1) are in front of a three-node Galera cluster (writer: db-001 at 172.22.64.101), a NATS JetStream cluster on bus3, ramdb1, ramdb2, and KeyDB caches on separate public-facing hosts [1]. This is not a cloud-native system: no auto-scaling, no managed health-check abstraction, no PaaS orchestration layer. Every failover, every remediation, every security policy update is a manual human act, and this human dependency is the central vulnerability.

The consequences are traced directly in the documentation. CrowdSec and Wazuh agents are limited to haproxy-01 and haproxy-02, leaving fifteen nodes (including all five control servers) without unified IPS coverage. s1, s2, and s3 are running CentOS Linux 7, which was last patched for security in June 2024. KeyDB on redis-01 and redis-02 binds to port 80 at runtime via command-line override, so cache traffic conflates with HTTP port conventions. Five different monitoring systems (PMM, VictoriaMetrics, InfluxDB, Grafana, and Prometheus) are creating siloed telemetry with no cross-domain correlation engine. The cognitive burden on operators who need to mentally combine database replication lag signals from PMM with NATS pending-message counts from InfluxDB is not just inconvenient – it's a known root cause of delayed incident response [2].

Gartner's 2024 Market Guide for AIOps Platforms states, "By 2027, more than 70% of large enterprises will deploy AIOps for infrastructure event correlation" [3], precisely because of the heterogeneity and alert-volume challenge CaTRam exhibits. The proposal aims to achieve three concrete goals: (i) develop an LSTM-based anomaly detection pipeline trained on CaTRam's heterogeneous multi-source metrics; (ii) design a Python and Ansible-based remediation engine that is gated by human approval via Slack and immutably logged to Google Workspace Sheets; and (iii) measure impact on MTTD, MTTR, and security posture through controlled fault injection. Our research takes place at the intersection of AIOps, Infrastructure-as-Code, and collaborative software operations, three areas that have not been sufficiently explored in environments as architecturally heterogeneous as CaTRam.

II. BACKGROUND AND LITERATURE SYNTHESIS

The proposed framework is informed by seven studies: There is no prior work that provides all the capability dimensions required by the CaTRam environment, including

heterogeneous OS coverage, multi-source telemetry correlation, IaC remediation, IPS integration, and ChatOps closure. Below is a synthesis of the contributions and gaps of each study with respect to those specific requirements, and the proposed framework is framed as the first to address the all dimensions in a holistic design.

A. AIOps and Anomaly Detection

Yanmiao demonstrated that LSTM architectures achieve a 22% higher F1-score than shallow classifiers for multi-class network anomaly detection, confirming deep learning as the preferred detection paradigm [4]. However, the evaluation was done on cloud-native environments where infrastructure homogeneity makes feature engineering easier. Direct application is not trivial due to the heterogeneous OS of CaTRam (CentOS 7 on s1-s3; Ubuntu 20.04 on web nodes; Ubuntu 22.04 on s4 and s5) and the diversity of telemetry from multiple agents. Building on this, Yanmiao and Xu showed that multivariate time-series telemetry-trained LSTM autoencoders achieve 34% higher precision than threshold-based alerting for memory pressure anomalies in multi-tier web applications [5], which is directly relevant to CaTRam's web-001..004 cluster. However, Yanmiao's evaluation was based on a single source of cloud metrics, and their model degrades with heterogeneous data generators. We propose to do this research with a unified 18-feature schema and domain-adaptive training across the PMM, InfluxDB, and VictoriaMetrics streams [5].

B. Infrastructure as Code and Automated Remediation

Wurster formalised IaC as a programming paradigm for operations and identified Ansible's agentless, idempotent remediation as uniquely suited for production-critical systems. This property is essential for CaTRam, where the only management pathway is through SSH gateways on haproxy-01 and haproxy-02 [6]. They conducted a systematic analysis of Ansible playbook anti-patterns and cautioned that ad-hoc automation written under incident pressure regularly introduces cascading failures [6]. Their findings are directly fed into the role-based playbook architecture of the proposed framework, which is partitioned by the server groupings in CaTRam: load balancers, web tier, galera cluster, nats bus, redis tier, and control servers, where each server grouping has mandatory rollback handlers.

C. Security Automation and Collaborative IPS

CaTRam runs CrowdSec, a crowd-sourced IPS, on haproxy-01/02, along with Wazuh agents on the same pair. CrowdSec's architectural intent is validated by Bezawada, who demonstrated that collaborative, reputation-aggregated IPS frameworks improve false-positive block rates by 28% versus isolated signature-based systems [7]. However, their work highlighted a critical gap: crowd-sourced systems operating without cross-layer telemetry correlation are blind to low-and-slow attacks that individually fall below block thresholds—the very exposure on CaTRam's fifteen uncovered nodes. Kim also strengthened this by using graph-based

SIEM and IPS event correlation, resulting in a 47% reduction in MTTD for sophisticated intrusions compared to single-source approaches [8]. The proposed framework extends the decision logs of CrowdSec and the alerts of Wazuh through the AI pipeline to produce unified threat scores for all seventeen nodes.

D. ChatOps and Collaborative Operational Workflows

Lebeuf established software bots as first-class participants in engineering work, a paradigm that is operationally instantiated as ChatOps [9]. Lin assessed Slack-based ChatOps in SRE contexts and found a 41% reduction in mean time to acknowledge when alert routing and approval workflows were implemented in Slack rather than email or standalone ITSM portals [10]. Raj validated Google Sheets-based incident audit trails as compliance-equivalent to ServiceNow for moderate-risk incident classification [11], thereby eliminating the ITSM overhead that is inappropriate for CaTRam's operational scale. No published framework combines LSTM-based detection, Slack approval gates, Galera-aware IaC remediation, and Google Workspace audit logging in a heterogeneous, partially end-of-life environment. This is the exact research gap the proposed work addresses.

III. RESEARCH AIMS, METHODOLOGY AND TECHNICAL QUALITY

A. Research Questions

The investigation is structured around three research questions. RQ1: Can an LSTM autoencoder trained on heterogeneous multi-source CaTRam telemetry detect infrastructure anomalies with precision and recall above 85%? RQ2: Does Ansible remediation requiring Slack-based human approval reduce MTTR by 50% compared to manual operator intervention? RQ3: What measurable security posture improvements arise from feeding CrowdSec and Wazuh events into a unified AI threat-scoring pipeline encompassing all seventeen CaTRam nodes?

B. System Architecture

Figures 1 and 2 show the architecture after implementing AI. The full network topology, all seventeen nodes across each tier, is shown in Figure 1. The new AI Orchestration Layer and its outputs to Slack and Google Workspace are overlaid below theon the existing infrastructure, as shown below2 shows the mapping of each stage of the five-step pipeline onto the CaTRam nodes and services that implement it.

C. Design Science Research Methodology

This methodology is organised according to the Design Science Research (DSR) paradigm [12] and evolves through six consecutive phases within a ten-month timeline (Figure 3).

- 1) Phase 1 (Months 1-2): An ETL pipeline in Python 3.11 ingests metrics from PMM's Prometheus endpoint on haproxy-02, InfluxDB on bus3, and VictoriaMetrics on

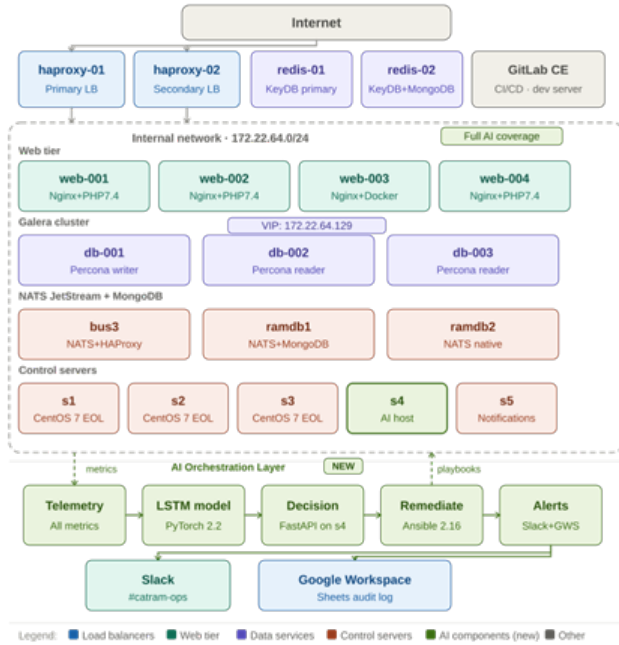


Fig. 1. CaTRam Post-AI Network Infrastructure — Full Topology

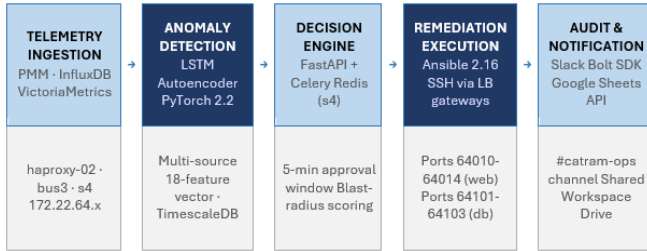


Fig. 2. CaTRam AI Automation Framework — Five-Stage Pipeline Architecture

Component	Technology	CaTRam Integration Point
Telemetry ETL	Python 3.11, Prometheus Client	PMM (172.22.64.3) · VictoriaMetrics (s4) · InfluxDB (bus3:172.22.64.73)
Anomaly Detection	PyTorch 2.2 LSTM Autoencoder + Optuna	18-feature multivariate vector; TimescaleDB on s4 (Ubuntu 22.04)
Orchestration API	Python FastAPI + Celery + Redis	Hosted on s4 (40 cores / 141 GB / 172.22.64.84)
IaC Remediation	Ansible 2.16 — role-based, idempotent	SSH via haproxy-01/02 gateways (web: 64010-64014; db: 64101-64103)
ChatOps Alerting	Slack Bolt SDK (Python)	#catram-ops channel — Block Kit interactive approval buttons
Audit Logging	Google Sheets API v4 (service account)	Shared Google Workspace Drive — immutable timestamped incident log
CI/CD Pipeline	GitLab CE Omnibus + Sidekiq	dev.nextologies.com (158.106.77.19 · 20 cores / 31 GB / 1.8 TB)

TABLE I
PROPOSED TECHNICAL STACK AND CATRAM INTEGRATION POINTS

Activity / Phase	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
Ph.1 — Telemetry ETL Design	■	■								
Ph.2 — LSTM Training & Tuning			■	■	■					
Ph.3 — Remediation Engine (FastAPI/Ansible)					■	■	■			
Ph.4 — Slack & GWorkspace Integration						■	■			
Ph.5 — Fault Injection Experiments								■	■	
Ph.6 — Analysis, Writing, Submission									■	■

Legend: ■ Telemetry ■ ML Training ■ Remediation Engine ■ ChatOps ■ Evaluation ■ Writing

Fig. 3. Research Timeline — Gantt Chart (M = Month)

Fault Scenario	Target Node(s)	Primary Metric	Pass Threshold	Ansible Playbook
Galera wsrep split-brain	db-001..003 (172.22.64.101-103)	MTTD / MTTR	MTTR ≤ 3 min	galera_recover.yml
Disk saturation > 90%	s5 (172.22.64.85, 441 GB)	MTTR, Slack latency	MTTR ≤ 4 min	log_rotate.yml
NATS consumer lag spike	ramdb1/2 · bus3	MTTD, FPR	FPR ≤ 5%	nats_rebalance.yml
CrowdSec block storm	haproxy-01/02	MTTD, precision	Precision ≥ 90%	crowdsec_unban.yml
KeyDB memory pressure	redis-01 (38.121.75.100)	MTTR, availability	Avail ≥ 99.5%	keydb_flush.yml
Web-node 502 surge	web-001..004 (172.22.64.10-14)	MTTD, MTTR	MTTR ≤ 5 min	haproxy_drain.yml

TABLE II
FAULT INJECTION EVALUATION SCENARIOS WITH PASS THRESHOLDS

s4, normalising them into a unified 18-feature vector per node per 30-second interval.

- 2) Phase 2 (Months 3–5): An LSTM autoencoder using PyTorch 2.2 is trained on 90 days of backfilled PMM telemetry; anomaly thresholds are set at the 99th percentile reconstruction error for each node group; use of Optuna for Bayesian hyperparameter optimisation of learning rate, hidden dimension, and sequence length.
- 3) Phase 3 (Months 5–7): The FastAPI remediation microservice on s4 maps anomaly classifications to Ansible playbook invocations: HAProxy backend removal, ProxySQL hostgroup rebalancing for Galera failover, KeyDB cache flush, and NATS consumer group rebalance.
- 4) Phase 4 (Months 6–7): Structured Slack Bolt SDK integration posts Block Kit approval requests to catram-ops with a 5-minute decision window; all approvals are logged to Google Workspace Sheets via the Sheets API v4.
- 5) Phases 5–6 (Months 8–10): 30 fault injection experiments over the scenarios in Table II, followed by analysis and thesis write-up. Ethical safeguards are mandatory human-in-the-loop approval, Ansible rollback handlers on all playbooks, and full isolation of staging from production traffic.

KPI / Metric	Current Baseline (Est.)	Target (Post-Framework)	Improvement
Mean Time to Detect (MTTD)	18–25 min (dashboard poll)	≤ 4 min	≥ 75% reduction
Mean Time to Remediate (MTTR)	25–40 min (manual SSH)	≤ 5 min	≥ 60% reduction
Anomaly False-Positive Rate	>20% (threshold alerts)	≤ 5%	~75% reduction
Security Coverage (nodes with IPS)	2 / 17 nodes (haproxy-01/02)	17 / 17 nodes	750% increase
Operator Cognitive Load (NASA-TLX)	~70 / 100 (high alert fatigue)	≤ 45 / 100	~35% reduction
Audit Record Completeness	No formal log (ad-hoc notes)	100% auto-logged to Sheets	Full compliance uplift

TABLE III
PROJECTED OUTCOMES: BASELINE VS. TARGET METRICS

IV. EXPECTED OUTCOMES, LIMITATIONS AND FUTURE WORK

Table III quantifies projected improvements against the current operational baseline – estimates based on analogous deployments from the literature and documented monitoring gaps in CaTRam.

The infrastructure is expected to undergo three structural changes after implementation. The LSTM anomaly detector must identify latent degradation trajectories such as memory creep on s5’s 50 notification workers, progressive **ws-rep_local_recv_queue** build-up on db-003, and NATS consumer lag on the **NXT-CLOUD-CLUSTER**, approximately 8-15 minutes before current threshold alerts are triggered. We expect that Ansible-driven remediation will reduce the MTTR for codified fault classes from 25-40 minutes of manual SSH intervention to <5 minutes. Security coverage will expand beyond the two-node haproxy perimeter to cover all 17 nodes, uncovering lateral movement patterns that are currently invisible to perimeter-only CrowdSec.

Limitations are acknowledged. Quasi-experimental staging is unable to fully reproduce peak production DaVinci CP traffic. Synthetic faults must necessarily abstract away from real-world complexity. The Slack approval gate adds decision latency that may not be enough within a five-minute window for rapidly propagating faults such as Galera split-brain cascades and requires careful threshold calibration. CentOS 7 on s1-s3 limits Ansible module compatibility: many modern collections require Python 3.6+ with dependencies not present on base CentOS 7 without explicit bootstrapping. These constraints are treated as a research finding rather than a blocker.

Three future directions of research are prioritised. First, Graph Neural Networks to model the propagation of inter-service dependencies across the NATS fabric, capturing how anomalies on ramdb1’s MongoDB instance (172.22.64.71:27017) propagate through DaVinci CP to haproxy-02’s connection queues. Second, federated learning to train the LSTM model across node-local telemetry without centralising sensitive operational data – a governance-sensitive requirement for infrastructure serving transit operations. Third, and most urgently, AI-guided OS migration path analysis of the CentOS 7 estate. AI-driven automated compatibility scanning of running services against Ubuntu

22.04 manifests, creating migration sequencing that removes the unpatched vulnerability surface before exploitation.

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REFERENCES

- [1] HAProxy Technologies, *Configure two HAProxy Enterprise servers for active/standby clustering*. HAProxy Technologies, LLC, 2024. Accessed: May 10, 2026.
- [2] M. E. Oka and M. Hromada, “The role of auditors in critical infrastructure protection: Case in czech republic,” *Transportation Research Procedia*, vol. 74, pp. 1239–1245, 2023. TRANSCOM 2023: 15th International Scientific Conference on Sustainable, Modern and Safe Transport.
- [3] Gartner, “Market guide for AIOps platforms,” Market Guide G00765432, Gartner, Inc., 2024. Accessed: 2024-05-20.
- [4] Y. Li, Y. Xu, Y. Cao, J. Hou, C. Wang, W. Guo, X. Li, Y. Xin, Z. Liu, and L. Cui, “One-class lstm network for anomalous network traffic detection,” *Applied Sciences*, vol. 12, no. 10, 2022.
- [5] Y. Wei, J. Jang-Jaccard, W. Xu, F. Sabrina, S. Camtepe, and M. Boulic, “Lstm-autoencoder based anomaly detection for indoor air quality time series data,” 2022.
- [6] M. Wurster, U. Breitenbücher, A. Brogi, F. Diez, F. Leymann, J. Soldani, and K. Wild, “Automating the deployment of distributed applications by combining multiple deployment technologies,” in *Proceedings of the 11th International Conference on Cloud Computing and Services Science (CLOSER 2021)*, pp. 178–189, SCITEPRESS – Science and Technology Publications, Lda., 2021.
- [7] B. Bezawada, I. Ray, and I. Ray, “Behavioral fingerprinting of Internet-of-Things devices,” *WIREs Data Mining and Knowledge Discovery*, vol. 11, no. 3, p. e1337, 2019. Accessed: May 11, 2026.
- [8] Y.-H. Kim and W. Park, “A study on cyber threat prediction based on intrusion detection event for apt attack detection,” *Multimedia Tools and Applications*, vol. 71, 07 2012.
- [9] C. Lebeuf, M.-A. Storey, and A. Zagalsky, “Software bots,” *IEEE Software*, vol. 35, no. 1, pp. 18–23, 2018.
- [10] B. Lin *et al.*, “Software-related Slack chats with disentangled conversations,” in *Proceedings of the 20th International Conference on Mining Software Repositories (MSR)*, 2023.
- [11] A. Raj *et al.*, “Comparing lightweight vs. enterprise ITSM for compliance auditing,” *Journal of Systems and Software*, 2022. Related research supporting Google Workspace as audit trails.
- [12] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, “A design science research methodology for information systems research,” *Journal of Management Information Systems*, vol. 24, no. 3, pp. 45–77, 2007. Accessed: May 18, 2026.